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19 August 1966
SUPERSEDING
FA-PD-MI-2361, Rev 1
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MILITARY SPECIFICATION
CARTRIDGE, INITIATOR, DELAY, M113

1. SCOPE

1.1 This specification covers Cartridge, Initiator, Delay, M113, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS
Munitions Command

D8596114	- Cartridge, Initiator, Delay, M113 Assembly
B8799620	- Box, Packing, Ammunition for Cartridge, Initiator, Delay, M70, M93, M113 or M155
B8799621	- Carton, Packing, Ammunition for Cartridge, Initiator, Delay M70, M93, M113 or M155
IEL-8596114	- Inspection Equipment List for Cartridge, Initiator, Delay, M113

FSC 1377

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2.1 (cont)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, assemblies, chemical and physical requirements, shall conform to the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Protective finish.- When protective finish is required by component details, finish requirements shall be as required on the drawing and referenced specification and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.

3.4 Delay element.- The Delay Element M13, shall conform to the applicable drawing and specification.

3.5 Vibration.- The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant, or derangement of any component.

3.6 Functional.

3.6.1 The cartridge shall not fail to fire.

3.6.2 Powder charges.- The igniter and propellant charges shall be fixed by the contractor to meet the specified ballistic requirements.

3.6.3 Function.- The cartridge tested in an M42 or M43 Initiator, shall comply with the requirements prescribed in table I.

Table I

Temperature	Pressure (psi) Min	Max	Delay Seconds (see 6.1)
-65° to -70° F	500	1500	4.2 max
160° to 165° F	500	1500	2.3 min

3.7 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

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3.7.1 Metal defects.- The metal part shall be free from cracks, splits, laminations, perforations, porosity or any similar defect.

3.7.2 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 30 cartridges and 10 sets of unassembled components, including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-414, as applicable.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of 600 cartridges, maximum, exclusive of test samples, and shall consist of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Primers from not more than one lot.

d. Delay element from not more than one lot.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in tables II and III. For an individual defect (single characteristic) the criteria in table II shall be used. For total defects within a major or minor class, the criteria in table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

Table II

Lot Size	Sample Size	Acceptance Numbers Major	Minor
1 - 300	100 % examination	0	0
301 - 1,300	110	1	2
1,301 - 3,200	150	2	3

Table III

Number of Defects in Class	Total Major Defects Allowable
2 - 3	1
4 - 7	2
8 - 19	3

Number of Defects in Class	Total Minor Defects Allowable
2 - 3	2
4 - 7	3
8 - 16	4

4.3.2.1 Classification of Defects.

4.3.2.1.1 Cartridge, Initiator, Delay, M113, Assembly, Dwg D8596114.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray/Visual 1/
2. Improper assembly	X-ray/Visual 1/
3. Mixed ammunition	Visual
4. Metal defective	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge profile failure	Gage
102. Diameter of body, max	Gage
103. Diameter of head, max	Gage
104. Length to datum diameter, max	Gage
105. O-ring split, cracked or cut	Visual

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Defect		Method of Inspection
106.	Protective finish inadequate or defective	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
201.	Total length	Gage
202.	Dent	Visual
203.	Draw scratch	Visual
204.	Fold, wrinkle, buckle or bulge	Visual
205.	Varnish inadequate or missing	Visual
206.	Surface finish improper	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.1.2 Cartridge Case, Dwg C8595202 covering a detail of Dwg D8596114.

CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Rockwell hardness incorrect	Gage
102.	Large diameter, max	Gage
103.	Small diameter, max	Gage
104.	Base wall thickness	Gage
105.	Wall thickness	Gage
106.	Length to datum diameter, max	Gage
107.	Concentricity of large and small diameters	Gage
MINOR		
201.	Total length, max	Gage
202.	Fold, wrinkle, buckle or bulge	Visual
203.	Surface finish improper	Visual
204.	Burr	Visual
205.	Overall length	Gage
206.	Radii incorrect	Gage/Visual
207.	Draw scratch	Visual

4.3.2.1.3 Disc, Sealing, Dwg C8594077 covering a detail of Dwg D8596114.

CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Diameter, max	Gage
102.	Thickness, max	Gage
103.	Marking missing, incorrect or illegible	Visual
MINOR		
201.	Dent	Visual

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4.3.2.1.3 (cont)

Defect	Method of Inspection
202. Varnish inadequate or missing	Visual
203. Surface finish improper	Visual

4.3.2.1.4 Carton, Packing, Dwg B8799621.

CRITICAL
None defined

MAJOR
None defined.

MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.1.5 Box, Packing, Dwg B8799620.

CRITICAL
None defined.

MAJOR
None defined.

MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Delay element examination.- Prior to assembly into a cartridge, each M13 Delay Element shall be visually examined for presence of primer and delay charge, or for deformation of any kind. Any deformation or the absence of the primer or delay charge shall be cause for rejection of the delay element.

4.3.4 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

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4.3.4.1 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.3.4.2 Airtightness.- Each cartridge assembly shall be subjected to the specified pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge assembly.

4.3.4.3 Vibration.- Twelve of the assembled cartridges of the initial production lot, or 12 assembled cartridges from a subsequent lot (when a change has been made in the lot of primers, propellant, or delay elements) shall be subjected to vibration as directed in 3.5. After vibration the 12 cartridges shall be x-rayed in accordance with Note 1/ of 4.3.2.1.1. Evidence of any disintegration of the primer, delay charge or propellant, or derangement of any component shall be cause for rejection of the lot.

4.3.4.4 Functioning.- This test will be performed by the Government. The 30 test cartridges, including the 12 vibrated cartridges of the initial production sample (or a subsequent production lot' sample of 30 cartridges when a change has been made in the lot of primers, propellant or delay elements, or the method of charging the delay elements) and 14 cartridges from each subsequent production lot, shall be grouped and temperature conditioned as prescribed in table IV and 4.4.3, and then tested in accordance with 4.4.4. Six of the 12 cartridges vibrated in accordance with 4.3.4.3 shall be included in each group tested over the temperature range prescribed in table IV under initial production lot.

Table IV

Temperature	Sample Size Initial Production Lot	Sample Size Production Lot	AQL
-65° to -70° F	15	7	0.65%
160° to 165° F	15	7	0.65%

4.3.4.4.1 The variables sampling plan of MIL-STD-414, example C2 or C3 as applicable, for the AQL listed in table IV shall be used. Failure of any cartridge to comply with the requirement of 3.6.3 or failure of any group to meet the criteria of the variables sampling plan, shall be cause for rejection of the lot.

4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8596114.

4.4 Test methods and procedures.

4.4.1 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL-8596114.

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4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Temperature conditioning.- The cartridge shall be loaded into an acceptable M42 or M43 Initiator and conditioned as prescribed in table IV. The temperature within the propellant shall be considered as the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a test device similar to the initiators undergoing tests. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

4.4.4 Test procedures.- Upon completion of the required temperature conditioning the cartridges will be ballistically tested in accordance with "Frankford Arsenal Manual of Functional Tests for Cartridge Actuated Devices for Aircraft Ejection Systems" (or such method as is approved by the contracting officer) within five minutes after removal from the conditioning chamber. The cartridges will be re-conditioned if this time is exceeded. When re-conditioning cartridges at minus 65 degrees to minus 70 degrees F, all condensation will be removed from the interior surface of the initiators before the initiators are returned to the conditioning chamber.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with drawings B8799620 and B8799621, except that marking shall reference M113 Cartridge.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure record.

Custodian:

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Preparing activity:

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Project No. 1377-A051

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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